

```
public class Queue<T>
{
    private Node<T> first;
    private Node<T> last;

    public Queue()
    {
        this.first = null;
        this.last = null;
    }

    public boolean isEmpty()
    {
        return this.first == null;
    }

    // adds to the end of the queue
    public void insert(T x)
    {
        Node<T> temp = new Node<T>(x);
        if (this.last == null)
        {
            this.first = temp;
        }
        else
        {
            this.last.setNext(temp);
        }
        this.last = temp;
    }

    // queue isnt empty
    // removes and returns the head
    public T remove()
    {
        T x = this.first.getValue();
        this.first = this.first.getNext();
        if (this.first == null)
        {
            this.last = null;
        }
        return x;
    }

    public T head()
    {
        return this.first.getValue();
    }

    public String toString()
    {
        String str = "[";
        Node<T> pos = this.first;
```

```
while(pos != null)
{
    if (pos.hasNext())
    {
        str = str + pos.getValue() + ",";
    }
    else
    {
        str = str + pos.getValue();
    }
    pos = pos.getNext();
}
str = str + "];
return str;
}
```